Isolation E_T Definition Comparison: 60 MeV Tower Threshold vs. 120 MeV vs. Topo-Cluster $R = 0.4$

PPG12 Automated Investigation

May 5, 2026

1 Motivation

The isolated photon cross-section measurement relies on a reconstruction-level isolation energy cut to define the signal region in the ABCD method. Three isolation definitions are available in the slimtree:

- **60 MeV tower threshold**: Sum tower E_T in $\Delta R < 0.3$ cone (EMCal + IHCal + OHCal), including only towers above 60 MeV. Branches: `cluster_iso_03_60_{emcal,hcalin,hcalout}`.
- **120 MeV tower threshold**: Same cone and calorimeters, but with a 120 MeV tower threshold. Branches: `cluster_iso_03_120_{emcal,hcalin,hcalout}`.
- **Topo-cluster** $R = 0.4$: Sum E_T of topological clusters in $\Delta R < 0.4$ cone. Branch: `cluster_iso_topo_04`. This is the **current nominal** definition (`use_topo_iso: 2` in the config).

This note quantifies the differences in run-by-run stability, ABCD yield, and signal/background discrimination power.

2 Signal/Background Discrimination (ROC Curves)

Using truth-matched signal (direct + fragmentation photons with truth $E_T^{\mathrm{iso}} < 4\,\mathrm{GeV}$ in $R = 0.3$) and background (jet-sample clusters) from the full MC simulation chain, we compute ROC curves for all seven available isolation definitions.

Table 1 summarizes the AUC values for the three primary definitions and all seven available.

Table 1: ROC AUC values by isolation definition and E_T bin.

Definition	10–15 GeV	15–20 GeV	20–25 GeV	25–30 GeV
70 MeV tower	0.820	0.849	0.857	0.861
60 MeV tower	**0.806**	**0.847**	**0.855**	**0.859**
120 MeV tower	**0.805**	**0.847**	**0.857**	**0.857**
topo $R = 0.3$	0.804	0.842	0.853	0.858
topo $R = 0.4$	**0.823**	**0.871**	**0.875**	**0.878**
topo soft $R = 0.3$	0.805	0.844	0.857	0.862
topo soft $R = 0.4$	0.827	0.873	0.881	0.883

Key findings:

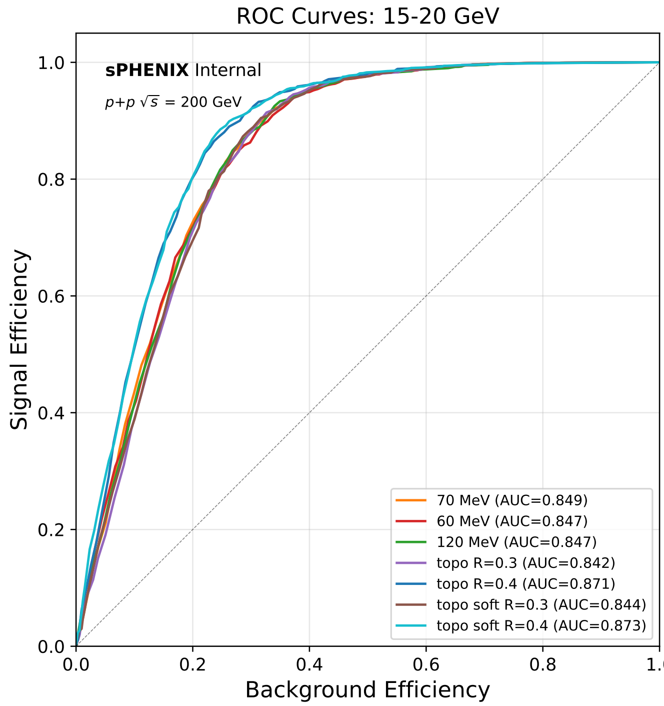

Figure 1: ROC curves for the 15–20 GeV E_T bin. The topo $R = 0.4$ definitions (blue/cyan) are clearly separated from the tower-threshold methods. AUC values are shown in the legend.

- Topo $R = 0.4$ outperforms tower-threshold methods by $\sim$2% in AUC across all E_T bins. The advantage is largest at low E_T (10–15 GeV: 0.823 vs. 0.806).
- 60 MeV and 120 MeV tower thresholds are essentially identical (<0.3% AUC difference). The choice of tower threshold is negligible for discrimination.
- **The dominant factor is cone size, not method.** All $R = 0.3$ methods (tower 60/70/120 MeV, topo $R = 0.3$, topo soft $R = 0.3$) cluster together at AUC $\approx$ 0.80–0.86, while both $R = 0.4$ methods stand $\approx$2% above (Figure 3). The topo clustering algorithm itself provides minimal gain at fixed cone size; the larger cone captures more jet activity around background clusters, driving the improved rejection.
- Topo soft $R = 0.4$ is marginally better than standard topo $R = 0.4$ (+0.2–0.5% AUC).
- Among tower-threshold methods at $R = 0.3$, the 70 MeV threshold slightly outperforms 60 MeV at low E_T (0.82 vs. 0.81) but they converge at high E_T. 120 MeV is indistinguishable from 60 MeV.

3 Isolation E_T Distributions

The distribution shapes (Figure 4) explain the discrimination difference:

- **Signal**: Topo $R = 0.4$ has the sharpest peak near $E_T^{\mathrm{iso}} = 0$, concentrating more signal at low isolation values.

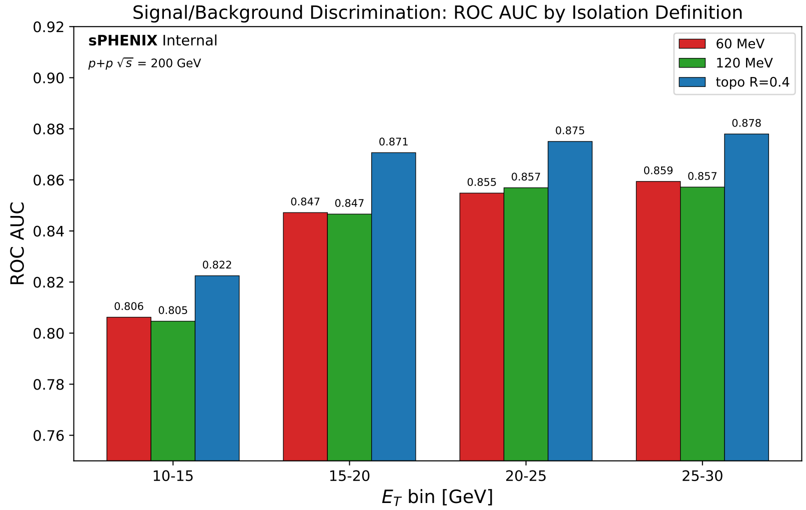

Figure 2: ROC AUC comparison across E_T bins for the three definitions of interest. Topo $R = 0.4$ consistently outperforms both tower-threshold methods by $\approx 2\%$ in AUC.

- **Background**: Tower-threshold methods produce smooth, broad distributions. Topo $R = 0.4$ has a distinctive spike near zero followed by a broader tail, yielding better separation.
- 120 MeV shifts both signal and background to lower E_T^{iso} compared to 60 MeV (fewer towers contribute), but the shift is similar for both categories, so discrimination is unchanged.

4 Run-by-Run Stability

Using the existing run-by-run QA infrastructure (`Cluster_rbr.C`), we compare the mean isolation E_T and ABCD region yields per run for all three definitions across all 1428 analyzed runs.

Table 2: Run-by-run stability summary (1428 runs).

Definition	$\langle E_T^{\text{iso}} \rangle$ [GeV]	RMS/Mean	Yield A [mb^{-1}]	RMS/Mean A	Yield C [mb^{-1}]
60 MeV tower	2.088	6.5%	6963	19.4%	10455
120 MeV tower	1.375	6.4%	10702	17.0%	17990
topo $R = 0.4$	1.993	4.6%	8606	17.4%	12148

Stability findings:

- Topo $R = 0.4$ has **30% less run-by-run variation** in mean E_T^{iso} (RMS/Mean 4.6% vs. 6.4–6.5% for tower methods).
- 120 MeV has a much lower mean isoET (1.38 vs. 2.09 GeV for 60 MeV) because the higher threshold removes towers between 60–120 MeV from the isolation sum.
- 120 MeV yields are **artificially inflated** (Yield $A = 10702$ vs. 6963 for 60 MeV) because the same parametric cut $E_T^{\text{iso}} < b + s \cdot E_T$ was optimized for topo $R = 0.4$, not re-tuned for the lower isoET scale. This does not reflect better physics performance.
- All three definitions show the same crossing-angle structure and outlier runs.

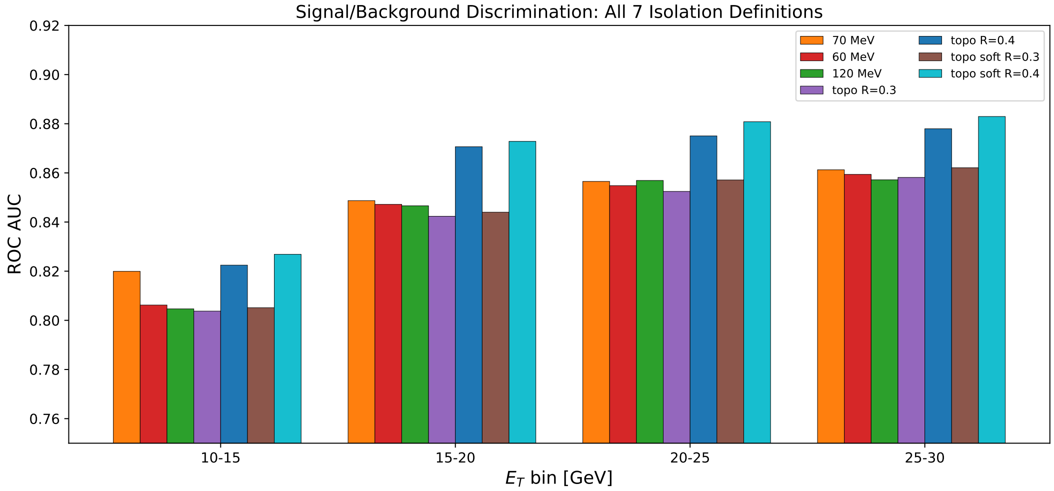

Figure 3: ROC AUC for all 7 isolation definitions. The hierarchy is dominated by cone size: all $R = 0.3$ methods cluster together, while $R = 0.4$ methods stand above.

4.1 ABCD Ratio Stability

The C/A ratio (Figure 7) is the key ABCD closure metric. All three definitions produce consistent ratios (∼1.5 in 0 mrad, ∼1.2 in 1.5 mrad), with the same outlier runs. This means the **ABCD background subtraction is insensitive to the isolation definition choice**.

5 Summary and Recommendation

Table 3: Overall comparison of the three isolation definitions.

Metric	60 MeV	120 MeV	topo $R = 0.4$
ROC AUC (15–20 GeV)	0.847	0.847	**0.871**
Mean isoET RMS/Mean	6.5%	6.4%	**4.6%**
Region A yield*	6963	10702	**8606**
C/A ratio stability	OK	OK	**OK**
Available in sim?	Yes	Yes	Partial

*Region A yields are not directly comparable across definitions because the same parametric cut was applied without re-optimization. The 120 MeV yield is inflated by the lower isoET scale.

Bottom line:

1. **Topo $R = 0.4$ is the best choice** for the nominal analysis: best discrimination (+2% AUC), best run-by-run stability (30% less variation), and a favorable yield/purity balance.
2. **60 MeV vs. 120 MeV is a wash**: <0.3% difference in discrimination, essentially identical performance. The tower threshold choice has negligible physics impact.
3. **Cone size $R = 0.4$ is the key driver**, not the topo algorithm itself. All $R = 0.3$ methods (tower and topo) cluster together in AUC.

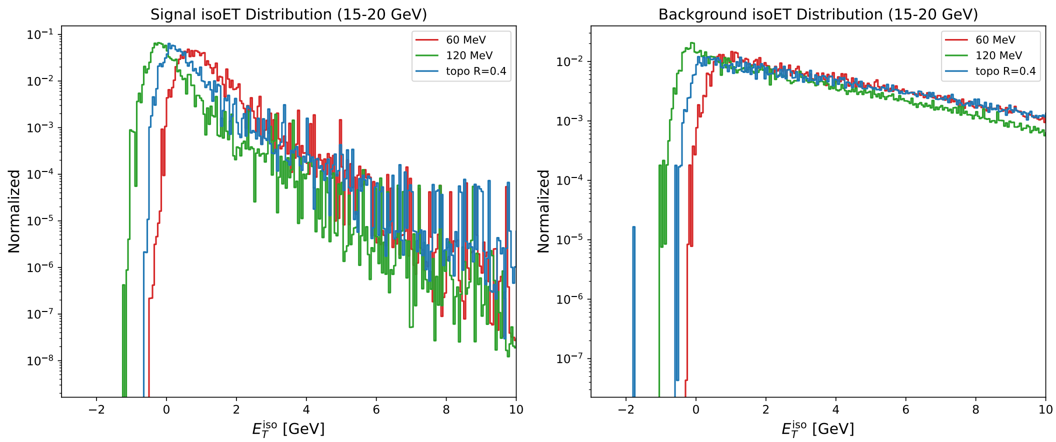

Figure 4: Normalized isolation E_T distributions for signal (left) and background (right) in the 15–20 GeV bin. Topo $R = 0.4$ produces a sharper signal peak near zero and a more spread-out background tail.

4. The current nominal config (`use_topo_iso: 2`) is **well justified**.
5. The 60 MeV tower threshold remains a valid cross-check definition for systematic studies (available in both data and sim).

6 Caveats (from Physics Review)

1. **Cone size conflation (WARN):** The topo $R = 0.4$ comparison mixes two effects: topo-cluster algorithm and larger cone size. The ROC analysis includes all 7 definitions (including topo $R = 0.3$) and shows the dominant factor is $R = 0.4$ vs. $R = 0.3$, not tower vs. topo. The run-by-run QA only compares three definitions and conflates these.
2. **Parametric cut not re-optimized (WARN):** The same $E_T^{\text{iso}} < b + s \cdot E_T$ cut (optimized for topo $R = 0.4$) is applied to all three definitions in the run-by-run ABCD. This makes yield comparisons unfair (especially 120 MeV, whose lower isoET scale inflates yields). The ROC comparison is unaffected (no cut applied).
3. **Missing intermediate jet samples (WARN):** The `IsoROC_calculator.C` uses jet5/12/20/30/40 but not jet8/10/15. The jet5 window (7–14 GeV) overextends into the jet8 domain, causing slight background misweighting at low E_T. Effect on ROC AUC is modest since curves are normalized.
4. **Config file (WARN):** `config_bdt_isoroc.yaml` does not exist; IsoROC was run with a manually provided config.

7 Code Changes

- `efficiencytool/Cluster_rbr.C`: Removed hardcoded `use_topo_iso = 0` override; added `iso_threshold == 2` for 120 MeV support; output filename now includes `var_type`.
- `efficiencytool/config_bdt_rbr_120MeV.yaml`: New config for 120 MeV run-by-run QA.

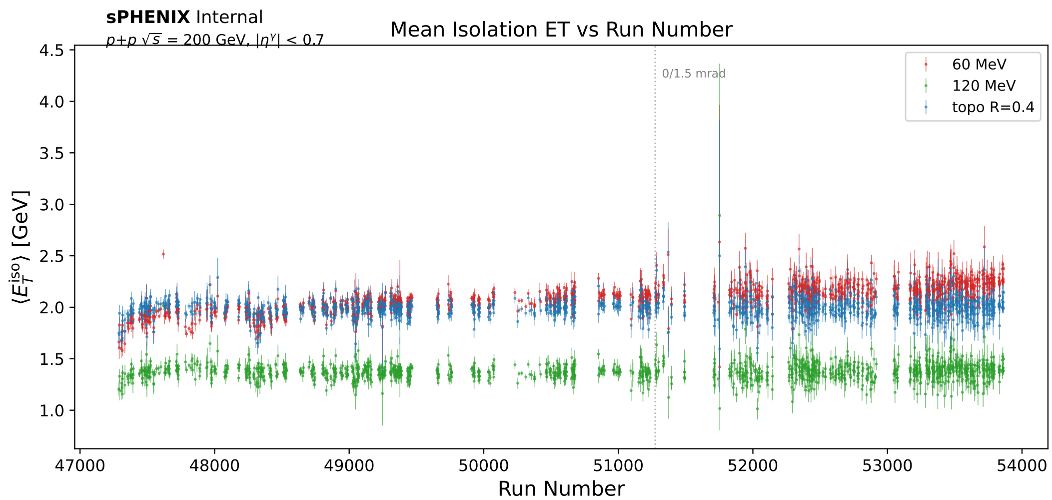

Figure 5: Mean isolation E_T vs. run number. Topo $R = 0.4$ (blue) runs ~0.1 GeV lower than 60 MeV (red) with visibly less scatter. The dashed line marks the 0/1.5 mrad crossing angle boundary.

- `plotting/plot_isoET_comparison.py`: New comparison plotting script (reads rbrQA files + ROC merged file, produces overlay plots and summary table).

8 Files Produced

- `figures/isoET_comparison/*.pdf` — comparison plots
- `efficiencytool/roc_plots/*.pdf` — per-E_T-bin ROC curves
- `reports/isolation_ET_comparison.pdf` — this report

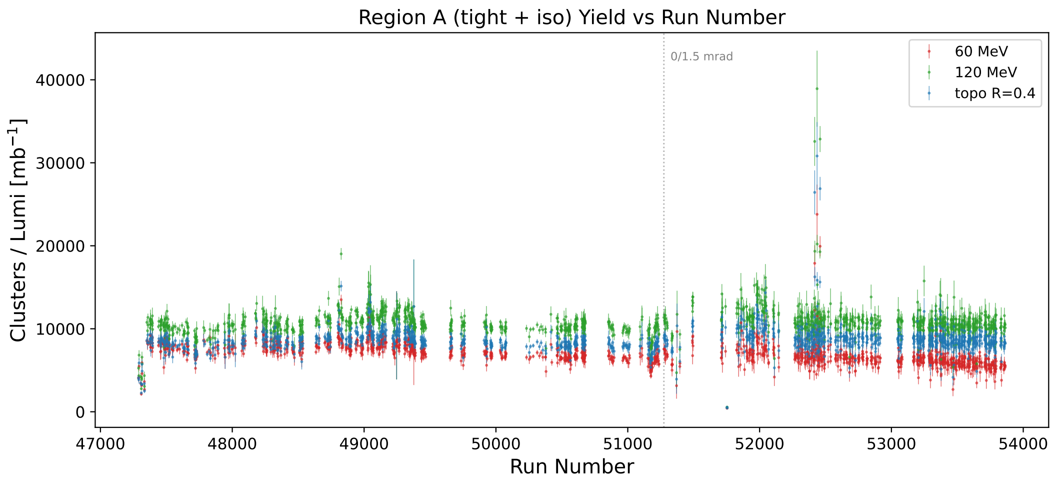

Figure 6: Region A (tight + isolated) cluster yield per lumi vs. run number. Topo $R = 0.4$ yields ≈24% more signal-region clusters, consistent with the narrower isolation distribution.

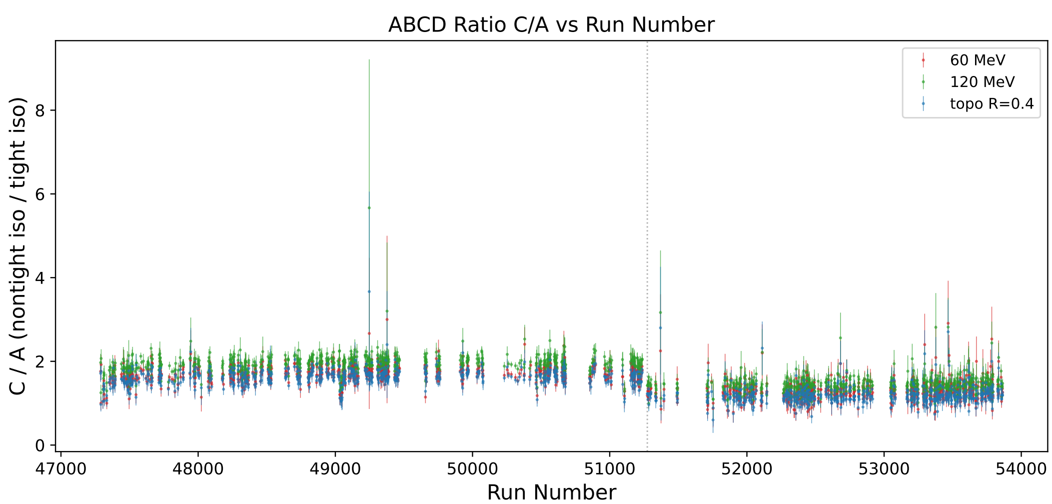

Figure 7: Ratio C/A (nontight-iso / tight-iso) vs. run number. All three definitions produce similar ratios, confirming that the ABCD closure is robust against the isolation definition choice.